

Proof Corrections

I found a few mistakes when reading this over. I'm sorry I didn't spot these earlier; they are all mistakes on my end. I've included a corrected version that fixes all these, but in case it's easier I thought I'd list them here.

Geometric Pooling Formula

On p. 252 (Introduction) and p. 261 (start of the Section 4 approximation discussion), the formula is set as

$$\sqrt{yz} / [\sqrt{yz} \cdot \sqrt{((1-y)(1-z))}]$$

The denominator should be a sum, not a product:

$$\sqrt{yz} / [\sqrt{yz} + \sqrt{((1-y)(1-z))}]$$

Search Committee Procedure

This is point 3 (p. 265), I messed up the weighted average formula. It currently reads:

Quinn's credence should be $0.8k + 32k/35$.

Both terms carry k . It should be:

Quinn's credence should be $0.8k + (32/35)(1-k)$.

The text around this is right, but the formula was wrong.

Incorrect subscript in equations (6) and (8) and in the Unanimity statement

Equation (6) (p. 254), equation (8) (p. 255), and the statement of Unanimity (p. 255) each set the second conjunct with $\mathbf{Cr}_A$ where it should be $\mathbf{Cr}_B$:

- Eq. (6): $\mathbf{Cr}_Q(p \mid \mathbf{Cr}_A(p) = y \wedge \mathbf{Cr}_A(p) = z) \rightarrow$ second conjunct should be $\mathbf{Cr}_B(p) = z$
- Eq. (8): same correction ($\mathbf{Cr}_B(p) = z$)
- Unanimity: $\mathbf{Cr}_Q(p \mid \mathbf{Cr}_A(p) = y \wedge \mathbf{Cr}_A(p) = y) = y \rightarrow$ second conjunct should be $\mathbf{Cr}_B(p) = y$

Equation (7) (p. 255) has it correct ($\mathbf{Cr}_A \wedge \mathbf{Cr}_B$), it's just there's a slip either side.

Related A/B slip in prose (p. 256)

When introducing B_x , the text reads "the disjunction of all possible evidence propositions that would lead **Ava** to have credence x in p ". Since this sentence defines B_x (Ben's evidence), it should read **Ben**.

Inline formula on p. 256 inconsistent with the proof

The text reads:

use the earlier derived fact that $\text{Cr}(A_x \mid \neg p) = ((1-x)r) / (x(1-r)\text{Cr}(p))$

This should be:

$$\text{Cr}(A_x \mid \neg p) = ((1-x)r) / (x(1-r)) \cdot \text{Cr}(A_x \mid p)$$

The version in the text has a spurious $\text{Cr}(p)$ in the denominator and omits the $\text{Cr}(A_x \mid p)$ factor. The preceding displayed proof and the substitution in the next display are both correct; only this prose restatement is wrong.

Wording corrections

- 251–252: “if Ava’s credence **in** y , and Ben’s is z ” $\rightarrow$ “if Ava’s credence **is** y , and Ben’s is z ”.
- 254: “approximating the value of (6) realistic cases” $\rightarrow$ “approximating the value of (6) **in** realistic cases”.
- 258: “The next model will help with more realistic case.” $\rightarrow$ “...with more realistic **cases**.”